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SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA04**

SLOT :

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25% Duration: 90 minutes

Assessment Tool Weightage: 25% Marks: 25

Code of Conduct:

I K.Neeha (192425245) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action**.**

Signature of the student: K.Neeha **Name of the student:K.Neeha**

Case Study Question 1: CNN for Automated Medical Image Classification (10 Marks)Assessment Overview

Component	**Details**	**Weightage**
Course	Deep Learning – MLA04	—
Course Outcome	CO3 – CNN, AlexNet, ResNet and Computer Vision	25%
Assessment Tool	CO3 Assessment Tool 1	25%
Duration	90 minutes	—
Questions	02	20 Marks

QUESTION 1 – MEDICAL IMAGE CLASSIFICATION

Scenario: Smart Chest X-Ray Screening System

A hospital is developing an automated screening system that receives chest X-ray images and predicts whether an image is Normal or Disease-Affected. Since X-rays may have different brightness, size and image quality, the model must learn robust visual patterns and avoid overfitting.

1. Proposed CNN Architecture

Stage	Layer / Operation	Purpose
1	Input: 224 × 224 × 1	Standardized grayscale X-ray
2	Conv 3×3, 32 filters + ReLU	Learns edges and simple patterns
3	Max Pooling 2×2	Reduces spatial size
4	Conv 3×3, 64 filters + ReLU	Learns textures and shapes
5	Max Pooling 2×2	Reduces computation
6	Conv 3×3, 128 filters + ReLU	Learns deeper abnormal patterns
7	Global Average Pooling	Converts feature maps to compact features
8	Dense 64 + Dropout	Combines learned features
9	Output: Sigmoid	Normal / Disease probability

CNN Flow Chart

2. Why CNN is Suitable

CNNs are suitable because they learn spatial patterns directly from images. Early layers detect edges and contours, middle layers learn textures and shapes, and deeper layers can capture medically relevant structures. This removes the need for manually designing image features.

CNN Component	Role in X-Ray System	Simple Example
Convolution	Extracts local visual features	Edges, lung boundaries
Filters	Detect specific patterns	Texture or abnormal region
ReLU	Adds non-linearity	Helps learn complex patterns
Pooling	Reduces feature-map size	Keeps important information
Fully Connected	Combines learned features	Final decision

Sigmoid	Produces probability	Disease = 0.87

3. Parameter Sharing

In a CNN, the same filter weights are reused across different image locations. Therefore, a 3×3 filter has only 9 weights plus bias, regardless of how many locations it scans. A fully connected network would require separate weights for every input-pixel connection. Parameter sharing therefore reduces memory, computation and the risk of overfitting while allowing the same pattern to be detected at different positions.

4. Regularization Strategy

Technique	Application	Benefit
Data Augmentation	Small rotations, shifts, brightness changes	Creates varied training examples
Dropout	Drop random neurons before final prediction	Reduces dependence on specific features
Batch Normalization	Normalize intermediate activations	Improves stable and faster training
Early Stopping	Stop when validation performance stops improving	Prevents unnecessary training
L2 Regularization	Penalize very large weights	Controls model complexity

5. AlexNet vs ResNet for Medical X-Rays

Factor	AlexNet	ResNet	Better Choice
Depth	Relatively shallow	Can be very deep	ResNet
Training	Older architecture	Residual connections ease training	ResNet
Feature learning	Good basic image features	Stronger hierarchical features	ResNet

Vanishing gradients	More difficult in deep networks	Skip connections help	ResNet
Computational cost	Lower in original form	Depends on version	AlexNet
Medical classification	Useful baseline	Usually stronger representation	ResNet

Decision

ResNet is the better choice for this application because its residual connections allow deeper feature learning while reducing the difficulty caused by vanishing gradients. A lightweight ResNet such as ResNet-18 can provide a practical balance between accuracy and computational cost. For a real clinical system, the model should also be validated carefully on diverse patient populations before deployment.

Result – Case Study 1

A CNN-based X-ray classifier was designed with convolution, pooling, feature extraction and classification stages. ResNet is recommended over AlexNet because it provides stronger deep feature learning and easier optimization.

QUESTION 2 – AUTONOMOUS VEHICLE OBJECT RECOGNITION

Scenario: RoadSense AI – Real-Time Vehicle Vision

An autonomous vehicle uses camera images to identify cars, pedestrians, bicycles, traffic signs and obstacles. The system must operate under sunlight, shadows, night conditions, different viewing angles and changing object distances.

1. CNN Design for Autonomous Driving

Stage	Operation	What the Model Learns
Input	Camera frame, resized and normalized	Raw road scene
Conv Block 1	Small filters + ReLU	Edges, corners and colour patterns
Pool 1	Downsampling	Important coarse information
Conv Block 2	More filters	Textures and simple shapes
Pool 2	Downsampling	Compact representation
Conv Block 3	Deep convolution	Vehicle/person/sign patterns
Global Pooling	Feature compression	High-level representation
Output Head	Softmax / detection head	Object class and confidence

2. Progressive Feature Learning

Network Depth	Typical Features	Road Example
Early layers	Edges and corners	Road-sign border
Middle layers	Textures and shapes	Wheel, face-like region
Deep layers	Object-level patterns	Car or pedestrian
Output layer	Class probabilities	Car = 0.94

The key idea is hierarchical learning: simple features learned in early layers are combined into more meaningful features in later layers. This allows the CNN to recognize an object even when its appearance changes due to distance, angle or lighting.

3. Parameter Sharing and Translation

A convolution filter scans the entire image using the same weights. If a car moves from the left side of the frame to the right side, the same learned filter can still detect relevant edges and textures. This weight reuse greatly reduces the number of parameters compared with connecting every pixel independently to every neuron.

4. Overfitting and Protection Methods

Risk	Example	Recommended Solution
Limited scene diversity	Model sees mostly daytime images	Day/night and weather augmentation
Memorization	Training accuracy very high	Dropout + weight regularization
Noise sensitivity	Blur or camera noise	Noise augmentation
Dataset imbalance	Few bicycle examples	Balanced sampling / class weights
Poor generalization	Works only on one road type	Diverse training data + validation

5. AlexNet vs ResNet – Autonomous Vehicle

Criteria	AlexNet	ResNet
Architecture	Early deep CNN with sequential layers	Deep CNN with residual/skip connections
Depth	Much shallower than modern ResNet variants	Can be very deep
Parameter efficiency	Original AlexNet has many parameters	Residual design supports deeper efficient variants
Training difficulty	Simpler but less suitable for very	Skip connections make deep

	deep learning	training easier
Vanishing gradient	More concern in deeper extensions	Residual paths help gradient flow
Feature learning	Strong historical baseline	Richer hierarchical representation
Real-time use	Fast enough in some settings	Choose lightweight ResNet variant for balance

6. Final Recommendation

ResNet is recommended for the autonomous vehicle vision system. Its residual connections support deeper networks and stronger feature extraction, which is valuable when objects appear in many forms. However, real-time performance depends on hardware and model size. A lightweight ResNet variant should be selected and optimized so that accuracy and inference speed are balanced.

Result – Case Study 2

A CNN pipeline was designed for road-object recognition. The model progressively learns low-level to high-level features, while parameter sharing reduces computation. ResNet is preferred because it provides stronger feature learning and better support for deep architectures.

Overall Analysis & Learning Summary

Concept	Medical X-Ray	Autonomous Vehicle
CNN	Detects image abnormalities	Recognizes road objects
Filters	Edges, textures, structures	Edges, signs, vehicles, people
Pooling	Reduces spatial dimensions	Reduces computation
Parameter Sharing	Fewer trainable weights	Fast scanning of scene
Regularization	Controls medical dataset overfitting	Improves robustness to road conditions
ResNet	Strong deep feature learning	Strong deep feature learning

Final Conclusion

CNNs are highly effective for image-based applications because they automatically learn hierarchical spatial features. Convolution filters detect local patterns, pooling reduces feature-map size, and parameter sharing makes CNNs much more efficient than fully connected image networks. Regularization improves generalization. Between AlexNet and ResNet, ResNet is generally the stronger modern choice for both medical image classification and autonomous vehicle vision, provided an appropriate lightweight variant is selected for computational constraints.

Faculty Evaluation

Criteria	Marks	Faculty Remarks
Conceptual Understanding	/ 2.5	
Linkages	/ 2.5	
Organization	/ 2.0	
Integration of Literature	/ 2.0	
Clarity & Presentation	/ 1.0	
Total	/ 10	

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand